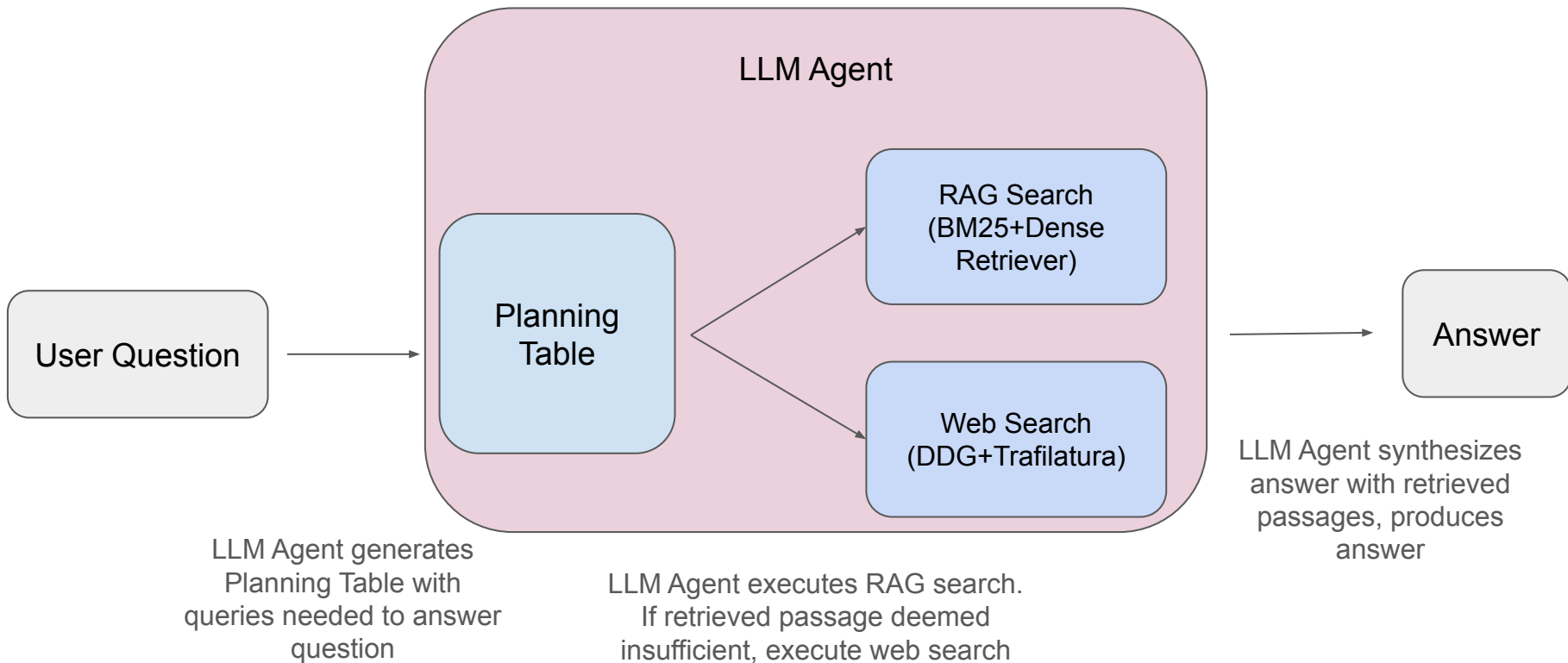


Agentic agent

Current status, progress, and issues

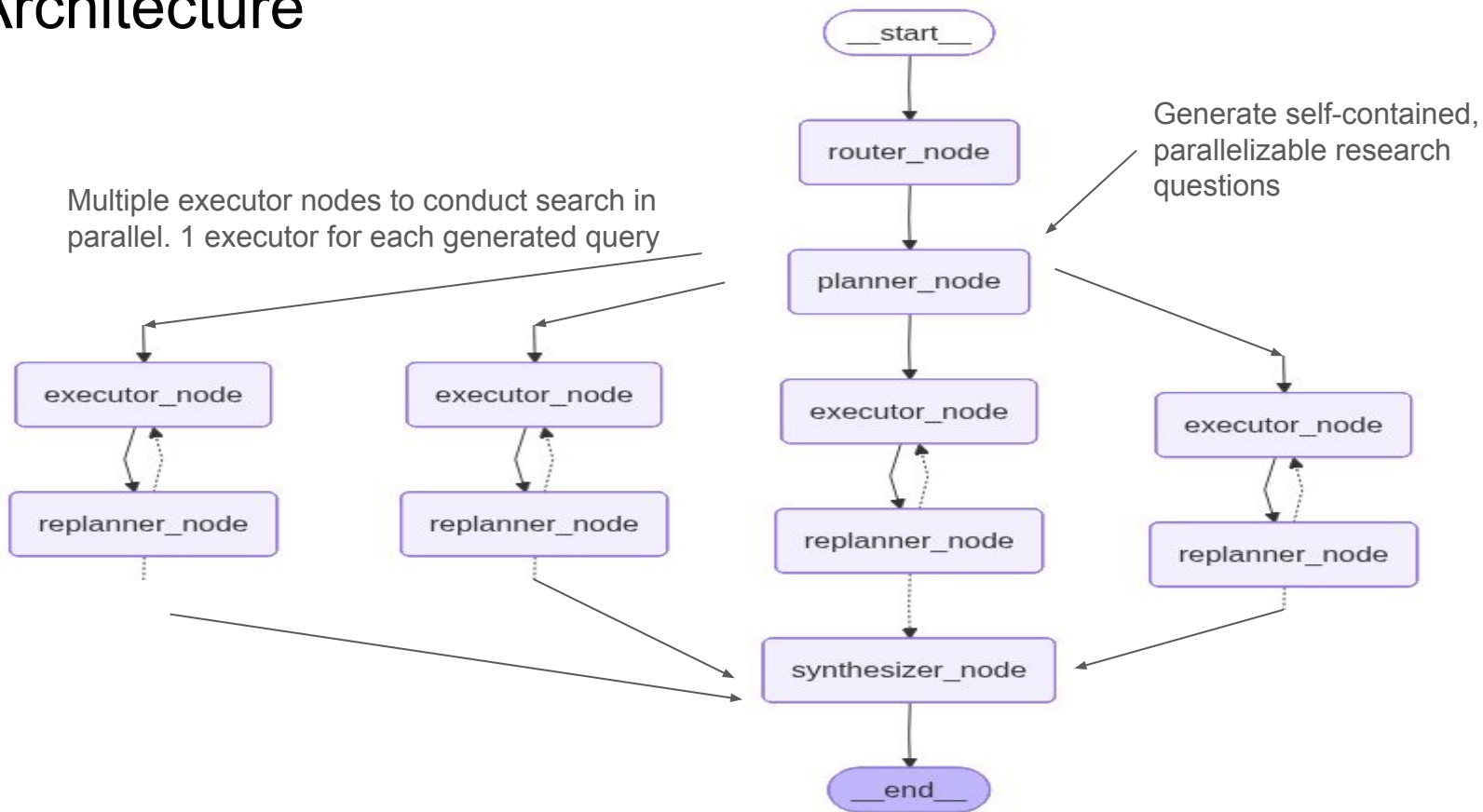
System Description



System Description

- **RAG Search:**
 - For each query (3 total), uses BM25 and Dense retriever to retrieve 15 passages each → produces 90 candidate passages
 - Deduplicates 90 candidate passages, then uses cross encoder to get top 5 passages from remaining candidate passages
- **Web Search:**
 - DuckDuckGo API to retrieve top 5 search results.
 - Contains brief description of source, not enough to answer question
 - Use web scraper to crawl to 5 URLs for more information
- Model used: deepseek-chat

Agent Architecture



Agent Architecture

- **Router Node:** Uses a LLM call to choose which ChromaDB collection(s) to search for the question (BarExam or Housing QA)
- **Planner Node:** Breaks the original question into 2-5 ordered legal research steps. Each step is self contained. It will determine whether to use RAG Search, Web Search, or directly answer
- **Executor Node:** Executes each research step using the selected strategy, then calls the judge LLM to decide whether the retrieval was sufficient. It stores the result, evidence, and confidence in state
- **Replanner Node:** Looks at the completed step from executor node and decides whether to next, retry, or complete.
 - If retrieval failed, it applies **deterministic** escalation: rag retry with re-written query → web search → direct answer
- **Synthesizer Node:** Combines all completed steps and the evidence index into the final IRAC-style answer with citations

Agent Skills Used by Nodes

- **Planner:** Break the question into 2-5 doctrine-focused sub-questions and choose rag_search, direct_answer, or web_search.
- **Query-rewrite:** Turn one sub-question into 1 primary + 2 alternate retrieval queries with dense legal keywords.
- **Judge:** Decide whether retrieved passages are sufficient, partial, or insufficient for the sub-question.
- **Replanner:** Decide whether to next, retry, or complete based on the original question and accumulated evidence.
- **Synthesizer:** Produce the final answer by combining all completed steps and evidence.

Benchmark overview

- Bar Exam QA: reasoning-heavy bar exam hypotheticals (multi-choice), retrieve supporting legal passages.
 - Questions sourced from historical Multistate Bar Exams. Gold passages curated by law students (explanation passage unavailable with historical MBEs)
 - 850k Passage Pool incl. Gold passages, 2019-2021 US caselaws, Wex, Cornell Law School
- Housing Statute QA: practical housing-law questions (Yes/No), retrieve supporting statutes by jurisdiction
 - Created from Legal Services Corporation Laws Database. Converts non Y/N answers into Y/N format for LLM eval

Bar Exam QA Dataset Sample

idx	dataset	example_id	prompt_id	source	subject	question_number
mbe_4	mbe	4	2	MBE-1972-78-part2		5

prompt	question	choice_a	choice_b	choice_c	choice_d	answer	gold_passage	gold_idx
Sue owned a five-acre tract of land, one acre of which had previously been owned by Opal, but to which Sue had acquired title by adverse possession. Sue contracted to convey the full five-acre tract to Peg, but the contract did not specify the quality of title Sue would convey. At closing, Peg refused the tendered deed and demanded return of her earnest money.	Suppose Peg pays the purchase price and accepts a deed. Subsequently, Sue's title to the one acre proves inadequate and Opal ejects Peg from that acre. Peg sues Sue for damages. Which of the following statements applies most accurately to the determination of Peg's rights?	Sue's deed was fraudulent.	The terms of the deed control Sue's liability.	The only remedy available for breach of warranty of title is rescission.	Peg's rights are based on the implied covenant that the title conveyed shall be marketable.	B	They are distinguishable from Schmidt & Wilson, 1972-1 F.2d 1018, 50-2 USTC ¶10,000, 28 AFTR2d 50-5811 (CA-9, 1950).	mbe_2318

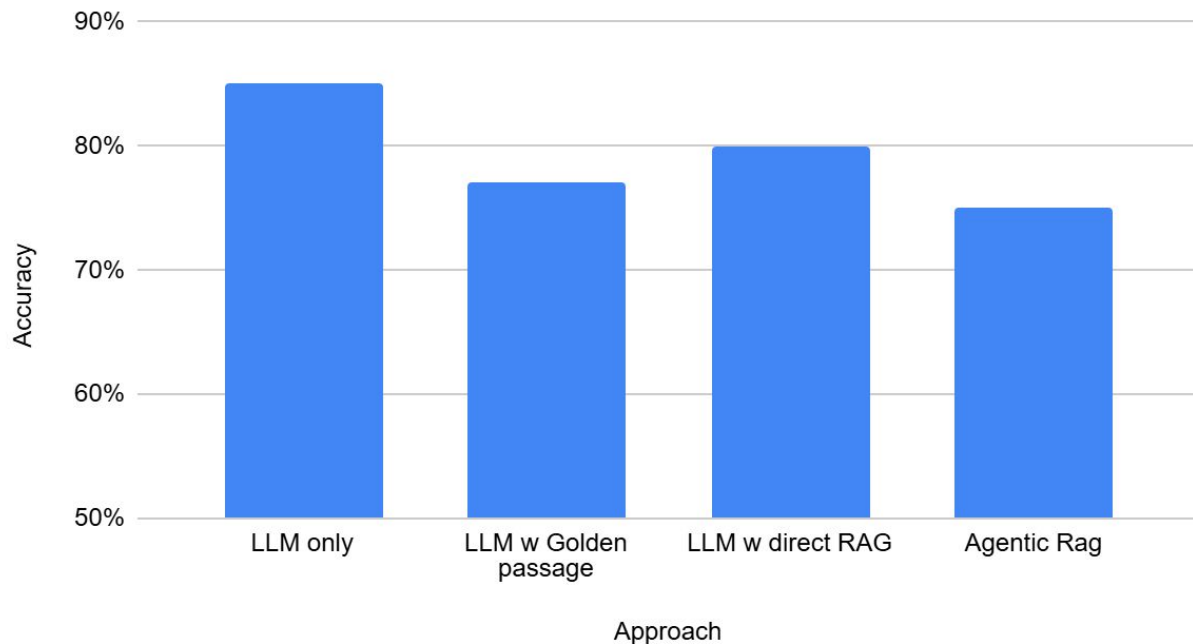
Benchmarks: Setup

100 question eval on BarExam-qa dataset, measure performance on no-RAG, simple-RAG, agentic-RAG using DeepSeek model

- LLM Only
 - Provide the LLM only with the question and multiple choice
- LLM with Golden Passage
 - Provide the LLM with the question, multichoice, and golden passage
- Direct RAG
 - LLM with vanilla RAG + query rewriting (primary query + 2 rewrites)
 - 2 alternative query are rewritten from primary query but using synonyms, related doctrines, alternative legal terms
 - Rag database: contains all embedded passages (600k) from BarExam dataset
- Agentic RAG (ours)

Benchmarks Performance (with DeepSeek v3.2)

100-q BarExam Eval with Different Approaches



LLM: deepseek-chat

100-q Eval Accuracies:

LLM-only: 85%

LLM with golden passage:
77%

LLM with direct RAG: 80%

Agentic RAG: 77%

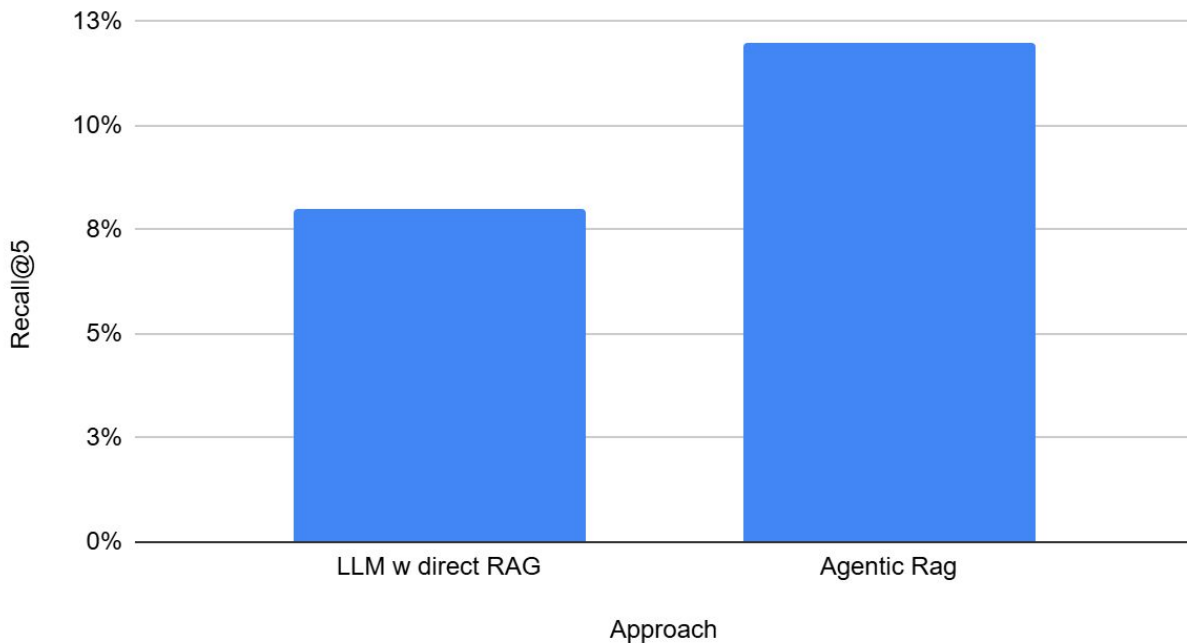
Benchmarks Performance - Golden Passage Retrieval

LLM: deepseek-chat

Recall@5 for gold
passage:

LLM with direct RAG: 8%
Agentic RAG: 15%

Direct vs Agentic RAG Recall@5 Performance



Synthesizer Skill Snippet

Task

Produce a cohesive, IRAC-structured legal answer that:

1. Directly addresses the original question
2. Grounds every claim in the provided evidence (no fabrication)
3. Maps citations to specific evidence entries using `[Evidence N]` format

Critical Rules

1. **Evidence-first, knowledge-supplemented.** Ground your answer primarily in the research findings and evidence index. Cite evidence as `[Evidence N]`. If the evidence is incomplete, you MAY supplement with established legal doctrine — but clearly distinguish: use `[Evidence N]` for passage-derived claims, and phrases like "Under established doctrine" for knowledge-based claims. Never fabricate specific statutes, dollar amounts, dates, or case names.
2. **Synthesize across steps.** Weave together findings from multiple research steps into a unified narrative. Do not just concatenate step results — integrate them coherently.
3. **Step attribution when useful.** When different steps addressed different aspects, briefly attribute: "Regarding the duty element [Evidence 1][Evidence 2]...; on causation [Evidence 4]..."
4. **Multiple-choice selection.** If the original question contains "Answer choices:", end your response with a blank line followed by: **Answer: (X)** selecting the letter best supported by your analysis. Use both the evidence and your legal knowledge to reason through the answer choices.

Judge Skill Snippet

Retrieval Sufficiency Judge

You are evaluating whether a set of retrieved passages (from a legal corpus or web search) is sufficient to support the sub-question's contribution to the original research goal.

Evaluation Criteria

Evaluate along four dimensions:

1. **Relevance**: Do the passages address the sub-question's legal doctrine, element, or rule? Passages that only tangentially relate to the topic (e.g., discuss a different doctrine using the same word) are not sufficient.
2. **Abstraction level**: Are the passages at the doctrine level (general legal rules, elements, definitions) rather than hyper-specific case facts that may not generalize? Doctrine-level passages are more useful for legal synthesis.
3. **Jurisdiction fit**: If the question is jurisdiction-specific, do the passages address the right jurisdiction? If jurisdiction is not specified, general common-law passages are acceptable.
4. **Completeness**: Do the passages, taken together, give enough information to make a grounded claim in the final answer? A partial answer is acceptable if combined with other completed steps — assess only whether this step contributed meaningfully.
5. **Direct support vs analogy**: If the passages discuss only adjacent doctrines, related concepts, or loose analogies, that is NOT sufficient. The retrieval must support the actual targeted doctrine, rule, element, current fact, or legal standard.

Output Format

Return ONLY valid JSON — no prose, no markdown fences:

```
```json
{
 "sufficient": "full",
 "reason": "One sentence explaining the verdict.",
 "missing": null,
 "suggested_rewrite": null
}
```
```

- `sufficient`: One of three values:

- `full` — the passages comprehensively address the sub-question.
 - `partial` — the passages address some aspects of the sub-question but are missing specific components. The evidence found IS useful and should be kept, but there are identifiable gaps.
 - `false` — the passages miss the key doctrine entirely or are too tangential to be useful.
- `reason`: Brief explanation tied to the specific sub-question and passages. Do not reference confidence scores.
- `missing`: If `sufficient` is `partial`, briefly state what specific information is missing (e.g., "missing the standard for breach" or "covers the rule but not the exceptions"). Set to `null` for `full` or `false`.
- `suggested_rewrite`: If `sufficient` is `false` and `action_type` is `rag_search`, provide an alternative query phrasing. Set to `null` otherwise.

Replanner Skill Snippet

Legal Research Replanner

You are deciding whether to continue, complete, or retry a research step, given that the judge has already determined the last step's retrieval was **sufficient**.

Your Decision

Choose one of three actions:

`next`

The last step produced useful evidence. Proceed to the next pending step.

Use when:

- The result substantively addresses the sub-question
- The accumulated evidence is growing toward a complete answer
- Pending steps are still needed to address aspects of the original question

`complete`

All necessary research has been gathered. Route to final synthesis.

Use when:

- All pending steps are done AND accumulated evidence can answer the original question
- Accumulated evidence already answers the original question (no need to execute remaining pending steps)
- Research has stalled despite escalation (judge has consistently found results insufficient across action types)

`retry`

Use this when the judge marked the last step sufficient, but you can see that the research direction is still wrong for the original question and a different phrasing would better serve the overall goal.

Only use `retry` when:

- The result is substantively off-topic relative to the original question (not just the sub-question)
- A more abstract or differently-worded query would clearly serve the research goal better

Retry limit enforced in code: If the step has already been retried once, the code will override `retry` with `next` or `complete`. Do not retry a step more than once.

When retrying, you MUST provide `revised\_question` — a rephrased version of the sub-question. Make it:

- More abstract (move from fact-specific → doctrine-level)
- Different terminology (if "exclusionary rule" failed, try "suppression of evidence Fourth Amendment")

Future steps/ongoing issues